Assignment 4

Problems:

1. Let
$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ -1 & 0 & 3 & 5 \\ 1 & -2 & 1 & 1 \end{bmatrix}.$$
Find a row-reduced echelon matrix R which is row-equivalent to A and an invertible 3×3 matrix P such that $R = PA$.

2. Let V be the set of all complex-valued functions f on the real line such that (for all t in $\mathbb{R}$)
$$f(-t) = \overline{f(t)}.$$
The bar denotes complex conjugation. Show that V, with the operations
$$(f+g)(t) = f(t) + g(t)$$
$$(cf)(t) = cf(t)$$
is a vector space over the field of real numbers. Give an example of a function in V which is not real-valued.

3. Let F be a field and let n be a positive integer ($n \geq 2$). Let V be the vector space of all $n \times n$ matrices over F. Which of the following sets of matrices A are subspaces of V?
   (a) all invertible A
   (b) all non-invertible A
   (c) all A such that $AB = BA$, where B is some fixed matrix in V
   (d) all A such that $A^2 = A$

4. Let W_1 and W_2 be subspaces of a vector space V such that the set-theoretic union of W_1 and W_2 is also a subspace. Prove that one of the spaces W_i is contained in the other.

5. Let W_1 and W_2 be subspaces of a vector space V such that $W_1 + W_2 = V$ and $W_1 \cap W_2 = \{\underline{0}\}$. Prove that for each vector α in V there are unique vectors α_1 in W_1 and α_2 in W_2 such that $\alpha = \alpha_1 + \alpha_2$.
Note: If $S_1, S_2, \ldots, S_k$ are subsets of a vector space V, the set of all sums
$$\alpha_1 + \alpha_2 + \cdots + \alpha_k$$
or vectors α_i in S_i is called the sum of the subsets $S_1, S_2, \ldots, S_k$ and is denoted by
$$S_1 + S_2 + \cdots + S_k$$
of by
$$\sum_{i=1}^{k} S_i.$$

6. Let V be a vector space over the complex numbers. Suppose that α, β, and γ are linearly independent vectors in V. Prove that $(\alpha + \beta)$, $(\beta + \gamma)$, and $(\gamma + \alpha)$ are linearly independent.